

for some vectors $\mathbf{u}$ and $\mathbf{v}$. If $\mathbf{u}$ is a unit vector $\mathbf{e}_i$, then (2.7.1) adds the components of $\mathbf{v}$ to the i th row. (Recall that $\mathbf{u} \otimes \mathbf{v}$ is a matrix whose i, j th element is the product of the i th component of $\mathbf{u}$ and the j th component of $\mathbf{v}$.) If $\mathbf{v}$ is a unit vector $\mathbf{e}_j$, then (2.7.1) adds the components of $\mathbf{u}$ to the j th column. If both $\mathbf{u}$ and $\mathbf{v}$ are proportional to unit vectors $\mathbf{e}_i$ and $\mathbf{e}_j$, respectively, then a term is added only to the element a_{ij} .

The *Sherman-Morrison* formula gives the inverse $(\mathbf{A} + \mathbf{u} \otimes \mathbf{v})^{-1}$ and is derived briefly as follows:

$$\begin{aligned}
 (\mathbf{A} + \mathbf{u} \otimes \mathbf{v})^{-1} &= (\mathbf{1} + \mathbf{A}^{-1} \cdot \mathbf{u} \otimes \mathbf{v})^{-1} \cdot \mathbf{A}^{-1} \\
 &= (\mathbf{1} - \mathbf{A}^{-1} \cdot \mathbf{u} \otimes \mathbf{v} + \mathbf{A}^{-1} \cdot \mathbf{u} \otimes \mathbf{v} \cdot \mathbf{A}^{-1} \cdot \mathbf{u} \otimes \mathbf{v} - \dots) \cdot \mathbf{A}^{-1} \\
 &= \mathbf{A}^{-1} - \mathbf{A}^{-1} \cdot \mathbf{u} \otimes \mathbf{v} \cdot \mathbf{A}^{-1} (1 - \lambda + \lambda^2 - \dots) \\
 &= \mathbf{A}^{-1} - \frac{(\mathbf{A}^{-1} \cdot \mathbf{u}) \otimes (\mathbf{v} \cdot \mathbf{A}^{-1})}{1 + \lambda}
 \end{aligned} \tag{2.7.2}$$

where

$$\lambda \equiv \mathbf{v} \cdot \mathbf{A}^{-1} \cdot \mathbf{u} \tag{2.7.3}$$

The second line of (2.7.2) is a formal power series expansion. In the third line, the associativity of outer and inner products is used to factor out the scalars λ .

The use of (2.7.2) is this: Given $\mathbf{A}^{-1}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$, we need only perform two matrix multiplications and a vector dot product,

$$\mathbf{z} \equiv \mathbf{A}^{-1} \cdot \mathbf{u} \quad \mathbf{w} \equiv (\mathbf{A}^{-1})^T \cdot \mathbf{v} \quad \lambda = \mathbf{v} \cdot \mathbf{z} \tag{2.7.4}$$

to get the desired change in the inverse

$$\mathbf{A}^{-1} \rightarrow \mathbf{A}^{-1} - \frac{\mathbf{z} \otimes \mathbf{w}}{1 + \lambda} \tag{2.7.5}$$

The whole procedure requires only $3N^2$ multiplies and a like number of adds (an even smaller number if $\mathbf{u}$ or $\mathbf{v}$ is a unit vector).

The Sherman-Morrison formula can be directly applied to a class of sparse problems. If you already have a fast way of calculating the inverse of $\mathbf{A}$ (e.g., a tridiagonal matrix or some other standard sparse form), then (2.7.4) – (2.7.5) allow you to build up to your related but more complicated form, adding for example a row or column at a time. Notice that you can apply the Sherman-Morrison formula more than once successively, using at each stage the most recent update of $\mathbf{A}^{-1}$ (equation 2.7.5). Of course, if you have to modify *every* row, then you are back to an N^3 method. The constant in front of the N^3 is only a few times worse than the better direct methods, but you have deprived yourself of the stabilizing advantages of pivoting — so be careful.

For some other sparse problems, the Sherman-Morrison formula cannot be directly applied for the simple reason that storage of the whole inverse matrix $\mathbf{A}^{-1}$ is not feasible. If you want to add only a single correction of the form $\mathbf{u} \otimes \mathbf{v}$ and solve the linear system

$$(\mathbf{A} + \mathbf{u} \otimes \mathbf{v}) \cdot \mathbf{x} = \mathbf{b} \tag{2.7.6}$$

then you proceed as follows. Using the fast method that is presumed available for the matrix $\mathbf{A}$, solve the two auxiliary problems

$$\mathbf{A} \cdot \mathbf{y} = \mathbf{b} \quad \mathbf{A} \cdot \mathbf{z} = \mathbf{u} \tag{2.7.7}$$

for the vectors $\mathbf{y}$ and $\mathbf{z}$. In terms of these,

$$\mathbf{x} = \mathbf{y} - \left[\frac{\mathbf{v} \cdot \mathbf{y}}{1 + (\mathbf{v} \cdot \mathbf{z})} \right] \mathbf{z} \quad (2.7.8)$$

as we see by multiplying (2.7.2) on the right by $\mathbf{b}$.

2.7.2 Cyclic Tridiagonal Systems

So-called *cyclic tridiagonal systems* occur quite frequently and are a good example of how to use the Sherman-Morrison formula in the manner just described. The equations have the form

$$\begin{bmatrix} b_0 & c_0 & 0 & \cdots & & & \beta \\ a_1 & b_1 & c_1 & \cdots & & & \\ & & \cdots & & & & \\ & & \cdots & a_{N-2} & b_{N-2} & c_{N-2} & \\ \alpha & & \cdots & 0 & a_{N-1} & b_{N-1} & \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ \cdots \\ x_{N-2} \\ x_{N-1} \end{bmatrix} = \begin{bmatrix} r_0 \\ r_1 \\ \cdots \\ r_{N-2} \\ r_{N-1} \end{bmatrix} \quad (2.7.9)$$

This is a tridiagonal system, except for the matrix elements α and β in the corners. Forms like this are typically generated by finite differencing differential equations with periodic boundary conditions (§20.4).

We use the Sherman-Morrison formula, treating the system as tridiagonal plus a correction. In the notation of equation (2.7.6), define vectors $\mathbf{u}$ and $\mathbf{v}$ to be

$$\mathbf{u} = \begin{bmatrix} \gamma \\ 0 \\ \vdots \\ 0 \\ \alpha \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ \beta/\gamma \end{bmatrix} \quad (2.7.10)$$

Here γ is arbitrary for the moment. Then the matrix $\mathbf{A}$ is the tridiagonal part of the matrix in (2.7.9), with two terms modified:

$$b'_0 = b_0 - \gamma, \quad b'_{N-1} = b_{N-1} - \alpha\beta/\gamma \quad (2.7.11)$$

We now solve equations (2.7.7) with the standard tridiagonal algorithm and then get the solution from equation (2.7.8).

The routine `cyclic` below implements this algorithm. We choose the arbitrary parameter $\gamma = -b_0$ to avoid loss of precision by subtraction in the first of equations (2.7.11). In the unlikely event that this causes loss of precision in the second of these equations, you can make a different choice.

```
void cyclic(VecDoub_I &a, VecDoub_I &b, VecDoub_I &c, const Doub alpha,      tridag.h
           const Doub beta, VecDoub_I &r, VecDoub_O &x)
Solves for a vector x[0..n-1] the "cyclic" set of linear equations given by equation (2.7.9). a,
b, c, and r are input vectors, all dimensioned as [0..n-1], while alpha and beta are the corner
entries in the matrix. The input is not modified.
{
    Int i,n=a.size();
    Doub fact,gamma;
    if (n <= 2) throw("n too small in cyclic");
```

```

VecDoub bb(n),u(n),z(n);
gamma = -b[0];
bb[0]=b[0]-gamma;
bb[n-1]=b[n-1]-alpha*beta/gamma;
for (i=1;i<n-1;i++) bb[i]=b[i];
tridag(a,bb,c,r,x);
u[0]=gamma;
u[n-1]=alpha;
for (i=1;i<n-1;i++) u[i]=0.0;
tridag(a,bb,c,u,z);
fact=(x[0]+beta*x[n-1]/gamma)/
      (1.0+z[0]+beta*z[n-1]/gamma);
for (i=0;i<n;i++) x[i] -= fact*z[i];
}

```

Avoid subtraction error in forming $bb[0]$.
Set up the diagonal of the modified tridiagonal system.

Solve $\mathbf{A} \cdot \mathbf{x} = \mathbf{r}$.
Set up the vector $\mathbf{u}$.

Solve $\mathbf{A} \cdot \mathbf{z} = \mathbf{u}$.
Form $\mathbf{v} \cdot \mathbf{x} / (1 + \mathbf{v} \cdot \mathbf{z})$.

Now get the solution vector $\mathbf{x}$.

2.7.3 Woodbury Formula

If you want to add more than a single correction term, then you cannot use (2.7.8) repeatedly, since without storing a new $\mathbf{A}^{-1}$ you will not be able to solve the auxiliary problems (2.7.7) efficiently after the first step. Instead, you need the *Woodbury formula*, which is the block-matrix version of the Sherman-Morrison formula,

$$\begin{aligned}
 & (\mathbf{A} + \mathbf{U} \cdot \mathbf{V}^T)^{-1} \\
 &= \mathbf{A}^{-1} - \left[\mathbf{A}^{-1} \cdot \mathbf{U} \cdot (\mathbf{1} + \mathbf{V}^T \cdot \mathbf{A}^{-1} \cdot \mathbf{U})^{-1} \cdot \mathbf{V}^T \cdot \mathbf{A}^{-1} \right]
 \end{aligned} \tag{2.7.12}$$

Here $\mathbf{A}$ is, as usual, an $N \times N$ matrix, while $\mathbf{U}$ and $\mathbf{V}$ are $N \times P$ matrices with $P < N$ and usually $P \ll N$. The inner piece of the correction term may become clearer if written as the tableau,

$$\left[\begin{array}{c} \mathbf{U} \end{array} \right] \cdot \left[\mathbf{1} + \mathbf{V}^T \cdot \mathbf{A}^{-1} \cdot \mathbf{U} \right]^{-1} \cdot \left[\begin{array}{c} \mathbf{V}^T \end{array} \right] \tag{2.7.13}$$

where you can see that the matrix whose inverse is needed is only $P \times P$ rather than $N \times N$.

The relation between the Woodbury formula and successive applications of the Sherman-Morrison formula is now clarified by noting that, if $\mathbf{U}$ is the matrix formed by columns out of the P vectors $\mathbf{u}_0, \dots, \mathbf{u}_{P-1}$, and $\mathbf{V}$ is the matrix formed by columns out of the P vectors $\mathbf{v}_0, \dots, \mathbf{v}_{P-1}$,

$$\mathbf{U} \equiv \left[\begin{array}{c} \mathbf{u}_0 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{u}_{P-1} \end{array} \right] \quad \mathbf{V} \equiv \left[\begin{array}{c} \mathbf{v}_0 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{v}_{P-1} \end{array} \right] \tag{2.7.14}$$

then two ways of expressing the same correction to $\mathbf{A}$ are

$$\left(\mathbf{A} + \sum_{k=0}^{P-1} \mathbf{u}_k \otimes \mathbf{v}_k \right) = (\mathbf{A} + \mathbf{U} \cdot \mathbf{V}^T) \tag{2.7.15}$$

(Note that the subscripts on $\mathbf{u}$ and $\mathbf{v}$ do *not* denote components, but rather distinguish the different column vectors.)

Equation (2.7.15) reveals that, if you have $\mathbf{A}^{-1}$ in storage, then you can either make the P corrections in one fell swoop by using (2.7.12) and inverting a $P \times P$ matrix, or else make them by applying (2.7.5) P successive times.

If you don't have storage for $\mathbf{A}^{-1}$, then you *must* use (2.7.12) in the following way: To solve the linear equation

$$\left(\mathbf{A} + \sum_{k=0}^{P-1} \mathbf{u}_k \otimes \mathbf{v}_k \right) \cdot \mathbf{x} = \mathbf{b} \quad (2.7.16)$$

first solve the P auxiliary problems

$$\begin{aligned} \mathbf{A} \cdot \mathbf{z}_0 &= \mathbf{u}_0 \\ \mathbf{A} \cdot \mathbf{z}_1 &= \mathbf{u}_1 \\ &\dots \\ \mathbf{A} \cdot \mathbf{z}_{P-1} &= \mathbf{u}_{P-1} \end{aligned} \quad (2.7.17)$$

and construct the matrix $\mathbf{Z}$ by columns from the $\mathbf{z}$'s obtained,

$$\mathbf{Z} \equiv \begin{bmatrix} \mathbf{z}_0 \\ \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_{P-1} \end{bmatrix} \quad (2.7.18)$$

Next, do the $P \times P$ matrix inversion

$$\mathbf{H} \equiv (\mathbf{I} + \mathbf{V}^T \cdot \mathbf{Z})^{-1} \quad (2.7.19)$$

Finally, solve the one further auxiliary problem

$$\mathbf{A} \cdot \mathbf{y} = \mathbf{b} \quad (2.7.20)$$

In terms of these quantities, the solution is given by

$$\mathbf{x} = \mathbf{y} - \mathbf{Z} \cdot \left[\mathbf{H} \cdot (\mathbf{V}^T \cdot \mathbf{y}) \right] \quad (2.7.21)$$

2.7.4 Inversion by Partitioning

Once in a while, you will encounter a matrix (not even necessarily sparse) that can be inverted efficiently by partitioning. Suppose that the $N \times N$ matrix $\mathbf{A}$ is partitioned into

$$\mathbf{A} = \begin{bmatrix} \mathbf{P} & \mathbf{Q} \\ \mathbf{R} & \mathbf{S} \end{bmatrix} \quad (2.7.22)$$

where $\mathbf{P}$ and $\mathbf{S}$ are square matrices of size $p \times p$ and $s \times s$, respectively ($p + s = N$). The matrices $\mathbf{Q}$ and $\mathbf{R}$ are not necessarily square and have sizes $p \times s$ and $s \times p$, respectively.

If the inverse of $\mathbf{A}$ is partitioned in the same manner,

$$\mathbf{A}^{-1} = \begin{bmatrix} \tilde{\mathbf{P}} & \tilde{\mathbf{Q}} \\ \tilde{\mathbf{R}} & \tilde{\mathbf{S}} \end{bmatrix} \quad (2.7.23)$$